

CASE STUDY ISE ACTIVITY – ANSWER SHEET

Course: Database Management System

Class: SY BCA

Duration: 2 Hours

Student Name: _____ **Roll No.:** _____

Case Study Title: Designing a Reliable and Scalable Database for an E-Commerce Platform

Read the case carefully and answer the following tasks in the space provided below.

Task 1: Transaction Management

Identify possible ACID property violations:

Suggest how to ensure atomicity and isolation using locks or 2PL:

Draw schedule diagram and check serializability:

Task 2: Indexing and Performance

Columns to be indexed:

Index type and justification:

Impact of improper indexing:

Task 3: NoSQL Integration

Design MongoDB JSON document for product catalog:

Explain BASE vs ACID difference:

Why choose NoSQL for scalability and flexibility:

Task 4: Database Security

Define three user roles:

Write GRANT and REVOKE SQL statements:

Suggest two database security measures:

Evaluation Rubric (For Teacher Use)

Criteria	Marks
Understanding of Concepts	5
Problem Analysis	5
Application & Design	5

Presentation & Justification	5
Total	20

Teacher's Signature: _____

Date: _____